

Decision Modelling (NPTEL Online Course 2021)

Tutorial 4 (Queuing Theory- L16-L20)

Detailed Answers

1. The service capacity of a queuing system can be optimized by:
 - i. Operating at maximum possible service capacity
 - ii. Minimizing the waiting costs only
 - iii. Minimizing the service cost only
 - iv. Minimizing both waiting and service cost together**

2. A sandwich maker has two counters. Each of these counters serve two different types of sandwiches. What kind of queueing system can form in front of the two counters? (Assume Poisson arrival and Exponential Service time).
 - i. One M/M/1
 - ii. Two M/M/1**
 - iii. One M/M/2
 - iv. Two M/M/2

3. In a small barber shop, only one customer can get hair cut while another customer can wait in a chair. Any other arriving customer has to wait outside, as there is only one chair available for waiting. The customers arrive randomly at 6 per hour. The service is exponential and takes 6 minutes on the average. Find the probability that an arriving customer will have to wait outside.
 - i. 24%
 - ii. 36 %**
 - iii. 40 %
 - iv. 60 %

Answer to Q3.

Average arrival rate $\lambda = 6$ per hour.

Average service time $1/\mu = 6$ minutes, and hence $\mu = 10/\text{hour}$

Hence, the utilization factor, $\rho = \lambda/\mu = 6/10 = 3/5$

The probability that an arrival will find the service free = $p(0) = 1 - \rho = 1 - (3/5) = 2/5$

The probability that an arrival will find the waiting chair free

$$= p(1) = \rho * p(0) = 3/5 * 2/5 = 6/25$$

So, Probability that an arriving customer will have to wait outside

$$= 1 - [2/5 + 6/25] = 9/25 = \mathbf{36\%}.$$

4. In a self-service restaurant, customer arrival is Poisson at 10 per hour. The service time is 3 minutes per customer (exponentially distributed). Find the average waiting time of a customer in the restaurant.

- i. **3 minutes**
- ii. 6 minutes
- iii. 9 minutes
- iv. 12 minutes

Answer to Q4.

Arrival rate $\lambda = 10/\text{hour}$ in the self-serving restaurant

Service time = 3 mins per customer = $3/60$ hours per customer

So, Service rate $\mu = 20/\text{hour}$

System utilization factor, $\rho = \lambda/\mu = 10/20 = 1/2$

Average No. of customers in the restaurant, $L = \rho = \frac{\lambda}{\mu} = 1/2$

Average waiting time in the restaurant: $W = \frac{L}{\lambda} = \frac{1}{20}$ hour = **3 minutes**

5. In a restaurant, arrival rate of customers is 15 per hour Poisson and Average Service Time is 4 minutes exponential in each counter. The restaurant serves two type of foods. Identify which option will keep the restaurant servers more busy.

- i. Opening two separate counters for each type of food
- ii. **Opening two counters with each of them serving both type of foods**
- iii. Opening a large number of self-service counters for each type of food
- iv. None of these

Answer to Q5.

Arrival rate $\lambda = 15$ per hour;

Service rate $\mu = 4$ mins per counter = 15 per hour per counter

Please see Lecture 18. Since $\lambda/\mu = 1$, for M/M/2 system, the busy period is 0.667

For self-service model, the busy period would be 0.632.

Hence, **Opening two counters with each of them serving both type of foods option would result in the queuing system becoming busier.**

6. Machines fail at 4 per hour and the cost of non-productive machine is Rs. 200 per hour. A repairman charges Rs. 100 per hour and repairs at 5 machines per hour. What will be the total queuing costs per hour? Assume M/M/1 queuing system.

- i. Rs. 300
- ii. Rs. 700
- iii. **Rs. 900**
- iv. Rs. 1000

Answer to Q6.

Arrival rate $\lambda = 4/\text{hour}$

Service rate $\mu = 5/\text{hour}$

System utilization factor, $\rho = \lambda/\mu = 4/5$

Average number of machines in repair, $L = \rho / (1 - \rho) = (4/5) / (1 - 4/5) = 4$

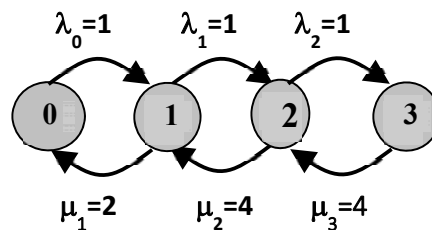
Cost of non-productive machine is Rs. 200 per hour

(A) Total cost of Non-productive machine is $200 * 4 = \text{Rs. } 800$

(B) Repairman's charge = Rs 100

Total Cost = (A) + (B) = **Rs. 900**

7. For the Rate diagram shown below, following relationship is true at Node 0:



- i. $P_0 = P_1$
- ii. **$P_0 = 2P_1$**
- iii. $2P_0 = P_1$
- iv. None of the above

Answer to Q7.

Please see the rate balance at Node 0. Hence we have $P_0 = 2P_1$.

8. Truck arrival rate is 15 per hour and the cost of automated unloading system is Rs. 100 per hour per truck. If unloading is done at the rate of 18 trucks per hour, what will be the total unloading costs per day, assuming 24-hour operations? Assume M/M/1 queuing system.

- i. Rs. 10000
- ii. Rs. 11000
- iii. **Rs. 12000**
- iv. Rs. 36000

Answer to Q8.

Arrival rate $\lambda = 15/\text{hour}$; Service rate $\mu = 18/\text{hour}$

System utilization factor, $\rho = \lambda/\mu = 15/18 = 5/6$

Average number of trucks in the unloading point, $L = \rho / (1 - \rho) = (5/6) / (1 - 5/6) = 5$

Cost of unloading is Rs. 100 per hour per truck, and the operation is for 24 hour

Total cost of unloading is $5 * 100 * 24 = \text{Rs. } 12000 \text{ per day}$

9. To visit a museum, rate of visitors is 20 per hour Poisson distributed. While the museum has a single ticket booking window, it takes 60 seconds to book a ticket (exponentially distributed). What is the average waiting time in the queue for a visitor to book a ticket?

- i. **30 sec**
- ii. 60 sec
- iii. 80 sec
- iv. 90 sec

Answer to Q9.

Arrival rate $\lambda = 20/\text{hour} = 2/6 \text{ per min} = 1/3 \text{ per min.}$

Service rate $\mu = 1 \text{ per min.}$

System utilization factor, $\rho = \lambda/\mu = (1/3)/1 = 1/3$

Average number of visitor in the system, $L = \rho / (1 - \rho) = (1/3)/(1-1/3) = 1/2$

Average number of visitor in the queue $L_q = L - \rho = 1/2 - 1/3 = 1/6$

So, Av. waiting time in the queue, $W_q = L_q / \lambda = (1/6) / (1/3) = 3/6 = 0.5 \text{ min} = 30 \text{ sec.}$

10. Cars arrive at an average rate of 8 per day to a service station having 12 hours operating time. The average time the mechanics require for servicing a car is about 30 minutes. If the arrival is Poisson and the service time is exponentially distributed, what is the utilization of the service station?

- i. 25%
- ii. 30%
- iii. **33%**
- iv. 42%

Answer to Q10.

Arrival rate $\lambda = 8/12 \text{ per hour} = 2/3 \text{ per hour.}$

Service time for one car = 30 minutes = $1/2 \text{ hour}$

Service rate $\mu = 2 \text{ per hour.}$

System utilization factor, $\rho = \lambda/\mu = (2/3)/(2) = 2/6 = 0.33 = 33\%$

11. In Question 10, if the arrival rate is reduced to 6 per day, how the utilization factor is affected?

- i. Increased by 6%
- ii. Decreased by 6%
- iii. Increased by 8%
- iv. **Decreased by 8%**

Answer to Q11.

Arrival rate $\lambda = 6/12$ per hour = $1/2$ per hour.

Service time for one car = 30 minutes = $1/2$ hour; Hence, Service rate $\mu = 2$ per hour.

System utilization factor, $\rho = \lambda/\mu = (1/2)/(2) = 1/4 = 0.25 = 25\%$

So the change in utilization is $25 - 33 = -8\%$

12. A small bank has two separate counters for deposit and withdrawal of money. Customer arrival is Poisson at 15 per hour to the deposit and 25 per hour to the withdrawal counter. Exponentially distributed service time in each counter is 2 minutes per customer. The average waiting time of a customer in the system for deposit counter and withdrawal counter is _____ and _____ respectively.

- i. 5 minutes, 10 minutes
- ii. **4 minutes, 12 minutes**
- iii. 4 minutes, 10 minutes
- iv. 5 minutes, 12 minutes

Answer to Q12.

For deposit

Arrival rate $\lambda = 15/\text{hour}$; Service time = 2 minutes per customer,

So, service rate $\mu = 1/2$ per min = 30 per hour

System utilization factor, $\rho = \lambda/\mu = 15/30 = 1/2$

Average number of customer in the system, $L = \rho / (1 - \rho) = (1/2)/(1-1/2) = 1$

Average waiting time, $W = L/\lambda = 1/15$ hour = **4 min**

For withdrawal

Arrival rate $\lambda = 25/\text{hour}$; Service time = 2 minutes per customer,

So, service rate $\mu = 1/2$ per min = 30 per hour

System utilization factor, $\rho = \lambda/\mu = 25/30 = 5/6$

Average number of customer in the system, $L = \rho / (1 - \rho) = (5/6)/(1-5/6) = 5$

Average waiting time, $W = L/\lambda = 5/25$ hour = **12 min**

13. The admission counter of a training center has deployed one person to collect the admission forms and fees for the new batch. It takes 90 seconds to complete the admission process for one student (exponentially distributed). The arrival rate of students is 35 per hour Poisson. If the training center wants to keep sufficient number of chairs for average number of students waiting in the queue, at least how many chairs would be required?

- i. 5
- ii. 6
- iii. **7**
- iv. 8

Answer to Q13.

Arrival rate $\lambda = 35/\text{hour}$; Service time = 90 seconds = $1/40$ hour per student,

So, service rate $\mu = 40$ per hour

System utilization factor, $\rho = \lambda/\mu = 35/40 = 7/8$

Average number of student in the system, $L = \rho / (1 - \rho) = (7/8) / (1 - 7/8) = 7$

Average no. students in queue $L_q = L - \rho = 7 - 7/8 = 6.125 = 7$ (Approx.);

So 7 chairs would be needed

14. Two quality control engineers are deployed to inspect a particular machine component arriving through a common incoming conveyor at the rate of 12 per hour. If the complete inspection of one component needs 7 minutes 30 seconds and the two engineers work on different pieces of the component, what should be the capacity of the workbench including the waiting components?

i. 3

ii. 4

iii. 5

iv. 6

Answer to Q14.

Arrival rate $\lambda = 12/\text{hour}$; Service time = 7.5 minutes = $1/8$ hour per component,

So, service rate $\mu = 8 \times 2 = 16$ per hour (since there are two engineers)

System utilization factor, $\rho = \lambda/\mu = 12/16 = 3/4$

Average number components in the system, $L = \rho / (1 - \rho) = (3/4) / (1 - 3/4) = 3$

15. A fast-food kiosk with a single server takes 5 minutes on an average to deliver the orders. If the customer arrival rate is 7 per hour, what is the average waiting time of the customers in the queue?

i. 5 min

ii. 7 min

iii. 12 min

iv. 15 min

Answer to Q15.

Arrival rate, $\lambda = 7/\text{hour} = 7/60$ per min.; Service time = 5 min; Service rate, $\mu = 1/5$ per min.

System utilization factor, $\rho = \lambda/\mu = (7/60) / (1/5) = 7/12$

Average number of customer in the system, $L = \rho / (1 - \rho) = (7/12) / (1 - 7/12) = 7/5$

Average number of customer in the queue, $L_q = L - \rho = 7/5 - 7/12 = 49/60$

Average waiting time in the queue $W_q = L_q / \lambda = (49/60) / (7/60) = 7/60$ hour = **7 min**

16. In a mine the excavators used are rented from a firm that charges Rs 2000 per hour. These specific type of excavators need routine maintenance and a separate team is employed for that at a cost of Rs. 550 per hour. If the rate at which excavators come for maintenance is 8 per hour, and the maintenance team has a servicing rate of 10 per hour, what will be the total queuing costs per hour? Assume M/M/1 queuing system.

- i. Rs. 7550
- ii. Rs. 8000
- iii. **Rs. 8550**
- iv. Rs. 9000

Answer to Q16

Arrival rate $\lambda = 8/\text{hour}$; Service rate $\mu = 10/\text{hour}$;

System utilization factor, $\rho = \lambda/\mu = 8/10 = 4/5$

Average number of excavator in maintenance, $L = \rho / (1 - \rho) = (4/5) / (1 - 4/5) = 4$

Rent of excavators is Rs. 2000 per hour

(A) Total cost of excavators in during maintenance is $2000 * 4 = \text{Rs. } 8000$ per hour

(B) Repairman's charge = Rs 550 per hour

Total Cost = (A) + (B) = **Rs. 8550 per hour**

17. A billing counter of a retail store having customer arrival rate at 10 per hour Poisson, installs a new billing machine that reduced billing time from 3 minutes per customer to 2 minutes per customer. What is the reduction in average waiting time for a customer in the system?

- i. 2 minutes
- ii. **3 minutes**
- iii. 3.5 minutes
- iv. 4 minutes

Answer to Q17.

Old machine

Arrival rate $\lambda = 10/\text{hour}$; Service time = 3 mins per customer = $3/60$ hours per customer

So, Service rate $\mu = 20/\text{hour}$

System utilization factor, $\rho = \lambda/\mu = 10/20 = 1/2$

Average No. of customers in the system, $L = \rho / (1 - \rho) = 1$

Average waiting time: $W_o = \frac{L}{\lambda} = \frac{1}{10}$ hour = 6 minutes

New machine

Arrival rate $\lambda = 10/\text{hour}$; Service time = 2 mins per customer = $2/60$ hours per customer

So, Service rate $\mu = 30/\text{hour}$

System utilization factor, $\rho = \lambda/\mu = 10/30 = 1/3$

Average No. of customers in the restaurant, $L = \rho/(1 - \rho) = 1/2$

Average waiting time in the restaurant: $W_n = \frac{L}{\lambda} = \frac{1}{20}$ hour = 3 minutes

Reduction in waiting time = $W_o - W_n = 6 - 3 = 3$ minutes

18. On weekdays, cars arrive at an average rate of 8 per day to a service station having 12 hours operating time. The average time the mechanics require for servicing a car is about 30 minutes. On weekends, the arrival of cars increases to 18 per day. If the arrival is Poisson and the service time is exponentially distributed, what is the change in utilization of the service station on weekends?

- i. Utilization increases by 30%
- ii. **Utilization increased 42%**
- iii. Utilization decreased by 30%
- iv. No change in utilization

Answer to Q18.

Weekdays

Arrival rate $\lambda = 8/12$ per hour = $2/3$ per hour.

Service time for one car = 30 minutes = $1/2$ hour; So, Service rate $\mu = 2$ per hour.

System utilization factor, $\rho = \lambda/\mu = (2/3)/(2) = 2/6 = 0.33 = 33\%$

Weekends

Arrival rate $\lambda = 18/12$ per hour = $3/2$ per hour.

Service time for one car = 30 minutes = $1/2$ hour; So, Service rate $\mu = 2$ per hour.

System utilization factor, $\rho = \lambda/\mu = (3/2)/(2) = 3/4 = 0.75 = 75\%$

So, the change in utilization is $75 - 33 = 42\%$ (increase in utilization)

19. In a manual ticket booking counter, the arrival rate of customers is 20 per hour Poisson distributed. The booking time is 60 seconds per ticket (exponentially distributed). If by installing an automated booking machine reduces the booking time to 40 seconds per ticket, what is the reduction in average waiting time in the system for the customers to book a ticket?

- i. 33.8 sec
- ii. 35.8 sec
- iii. **38.4 sec**
- iv. 40.2 sec

Answer to Q19.

Manual ticket booking

Arrival rate $\lambda = 20/\text{hour} = 2/6$ per min. Service time = 60 sec = 1 min.

Service rate $\mu = 1$ per min.

System utilization factor, $\rho = \lambda/\mu = (2/6)/1 = 2/6$

Average number of visitor in the system, $L = \rho / (1 - \rho) = (2/6)/(1-2/6) = 1/2$

Average waiting time, $W = L / \lambda = (1/2)/(2/6) = 1.5 \text{ min.} = 90 \text{ sec}$

Automated ticket booking

Arrival rate $\lambda = 20/\text{hour} = 2/6 \text{ per min.}$ Service time = 40 sec = 40/60 min. = 2/3 min

Service rate $\mu = 3/2 \text{ per min.}$

System utilization factor, $\rho = \lambda/\mu = (2/6)/(3/2) = 2/9$

Average number of visitor in the system, $L = \rho / (1 - \rho) = (2/9)/(1-2/9) = 2/7$

Average waiting time, $W = L / \lambda = (2/7)/(2/6) = 0.86 \text{ min.} = 51.6 \text{ sec}$

Reduction in waiting time = 90 – 51.6 = **38.4 sec.**

20. In a company, machines fail at 4 per hour and the cost of non-productive machine is Rs. 200 per hour. The existing repairman charges Rs. 100 per hour and repairs at 5 machines per hour. If a new repairman is to replace the existing one with a charge of 150 per hour and repair rate of 6 machines per hour, what will be the change in the total queuing costs per hour? Assume M/M/1 queuing system.

- i. Queuing cost will increase by Rs. 300 per hour
- ii. Queuing cost will decrease by Rs. 300 per hour
- iii. Queuing cost will increase by Rs. 350 per hour
- iv. Queuing cost will decrease by Rs. 350 per hour**

Answer to Q20.

Existing repairman

Arrival rate $\lambda = 4/\text{hour}$; Service rate $\mu = 5/\text{hour}$; System utilization factor, $\rho = \lambda/\mu = 4/5$

Average number of machines in repair, $L = \rho / (1 - \rho) = (4/5)/(1 - 4/5) = 4$

Cost of non-productive machine is Rs. 200 per hour

(A) Total cost of Non-productive machine is $200 \times 4 = \text{Rs. } 800 \text{ per hour}$

(B) Repairman's charge = Rs 100 per hour

Total Cost = (A) + (B) = Rs. 900 per hour

New repairman

Arrival rate $\lambda = 4/\text{hour}$; Service rate $\mu = 6/\text{hour}$; System utilization factor, $\rho = \lambda/\mu = 4/6$

Average number of machines in repair, $L = \rho / (1 - \rho) = (4/6)/(1-4/6) = 2$

Cost of non-productive machine is Rs. 200 per hour

(A) Total cost of Non-productive machine is $200 \times 2 = \text{Rs. } 400 \text{ per hour}$

(B) Repairman's charge = Rs 150 per hour

Total Cost = (A) + (B) = Rs. 550 per hour

Hence, Change in queuing cost per hour = 550 – 900 = **–350**